Reconstruction Is Not Replication: Cross-Validating Sparse Autoencoder Features on Pythia-160M

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Abstract

We train a sparse autoencoder (SAE) on the layer-6 residual stream of Pythia-160M and evaluate it along six independent quality axes, including three cross-validation tests that the field rarely reports together: cross-width consistency (training the same SAE at $N \in \{4096, 16384, 65536\}$ features), cross-seed replication (training at the same hyperparameters with a different random seed), and cross-architecture agreement (training a Matryoshka SAE [7] with nested-k bottlenecks). The reference SAE achieves 98.4% variance explained, 95.4% loss recovered under cross-entropy substitution (a competitive number for the small-model band), and 88.5% auto-interp monosemanticity rate over the 26 most-active features. The same SAE has only 19.9% of features at cosine similarity > 0.9 across a different random seed, and only 10.6% at the same threshold across the Matryoshka architecture. Combining all four cross-validation conditions, only 2.14% of features (351 of 16,384) survive at cos > 0.9. Of the 26 features that an automatic interpreter labeled as monosemantic, only 3 are fully stable: features encoding BibTeX/LaTeX citation references, file paths, and logical operators / negation. We argue that the standard suite of SAE quality metrics (reconstruction, sparsity, monosemanticity rate) systematically *overstates* feature reliability, by approximately 8× on this setup. We release a multi-metric scorecard tool and a per-feature stability navigator that surface this gap on any SAE.[1]

1 Introduction

Sparse autoencoders (SAEs) have become the dominant tool for converting a transformer's internal activations into a sparse, interpretable basis [1, 2]. The standard quality story has three parts: reconstruction (variance explained or MSE), sparsity (an L_0 target enforced by ReLU+L_1 [1], top-k [3], or related mechanisms [4, 5]), and interpretability (the fraction of recovered features that a human or LLM inspector labels as monosemantic). Most published SAE work reports these three.

This paper asks a question the standard suite cannot answer: if you train the same SAE again — with a different random seed, a different width, or a slightly different architectural objective — how many of the features it recovered are the same?

Our finding: *very few.* On Pythia-160M layer 6 with $N = 16{,}384$ features and $k = 64$:

- Variance explained is 98.4%; CE-delta loss recovered is 95.4%.
- Auto-interp on the top 26 features classifies 88.5% as monosemantic.

[1] Code: https://github.com/OE-GOD/sae-pythia-160m.

- But cross-seed cosine similarity is > 0.9 for only 19.9% of features; cross-Matryoshka is > 0.9 for only 10.6%.
- Of the auto-labeled features, only 3 of 26 are fully stable across width, seed, and architecture. The rest, including all 11 apparent variants of "newline tokens" that the SAE found, collapse under cross-validation.

These observations have two implications. First, when the field reports SAE quality as a single reconstruction or interpretability number, that number can overstate the reliability of recovered features by nearly an order of magnitude. Second, the small subset of features that does survive cross-validation is, plausibly, the right basis for downstream work — circuit discovery, causal interventions, agent behavior auditing — because those features are not artifacts of the particular SAE training procedure.

Contributions.

1. Empirical: a six-axis evaluation of a TopK SAE on Pythia-160M layer 6, including cross-width (§5.1), cross-seed (§5.2), and cross-architecture (§5.3) replication results. The cross-validation numbers are roughly $\sim 5\times$–$10\times$ worse than the reconstruction and interpretability numbers on the same SAE.
2. A multi-metric scorecard tool that reports all six axes side-by-side (§5.5). On our reference SAE the strongest axis is 0.984 and the weakest is 0.106, a ~ 88 percentage-point gap.
3. A per-feature stability navigator that reports a feature's best-match identity in every comparison SAE (§5.4). The output is a shortlist of 351 fully stable features (2.14% of the total) that we treat as the strongest "real feature" candidates in this setup.

Scope and honesty. This is a single-model, single-layer study on a small open transformer (160M params) with 1M training tokens and a $50 compute budget. We make no claims about frontier-scale models. We do claim that on this benchmark the gap between reconstruction-style and replication-style metrics is large enough to deserve attention in any SAE quality report.

2 Background

The SAE setup is standard. For an activation $h \in \mathbb{R}^d$ at a target layer,

$$f = \sigma(W_{\text{enc}}(h - b_{\text{dec}}) + b_{\text{enc}}), \quad f \in \mathbb{R}^N, \tag{1}$$

$$\hat{h} = W_{\text{dec}} f + b_{\text{dec}}, \tag{2}$$

with σ enforcing sparsity. We use TopK [3], which keeps the k largest pre-activations and zeros the rest. This gives $L_0 = k$ by construction and removes the L_1-shrinkage problem [4] we observed in an initial vanilla-L_1 run (Appendix A).

The Matryoshka SAE variant [7] introduces nested-k reconstruction bottlenecks: for $k_1 < k_2 < \cdots < k_M$, features active at smaller k_i are a strict subset of features at larger k_j, and the loss sums reconstruction over all levels:

$$\mathcal{L}_{\text{matryoshka}} = \sum_{i=1}^{M} \| h - W_{\text{dec}} f_{\text{topk}}^{(k_i)} - b_{\text{dec}} \|_2^2. \tag{3}$$

We use this variant for the cross-architecture comparison in §5.3.

3 Setup

Model and data. Pythia-160M [8] ($d_{\text{model}} = 768$, 12 layers), accessed via TransformerLens. Target: residual stream at the output of layer 6, hook `blocks.6.hook_resid_post`. Training corpus: `NeelNanda/pile-10k`, a 10k-document subset of the Pile. We tokenise the first $\sim 8{,}000$ documents into a flat stream and reshape into 7,812 sequences of length 128, giving 999,936 training tokens. Held-out evaluation tokens are drawn from documents indexed $\geq 8{,}001$.

SAE training. The reference SAE (referred to as the "16k SAE") has $N = 16{,}384$ features ($\sim 21\times$ expansion over d_{model}), $k = 64$ (TopK), b_{dec} initialised to the empirical activation mean, tied initialisation ($W_{\text{enc}} = W_{\text{dec}}^{\top}$), decoder column unit-norm constraint enforced after each Adam step, lr $= 10^{-3}$, batch size 4,096, 20,000 training steps. Three additional SAEs are trained for cross-validation:

- $N = 4{,}096$ (cross-width, smaller, §5.1)
- $N = 65{,}536$ (cross-width, larger, §5.1)
- Second random seed (`torch.manual_seed(42)`, all other hyperparameters identical, §5.2)
- Matryoshka SAE at $k_{\text{levels}} = \{16, 64, 256\}$ (§5.3)

Hardware and compute. Apple Silicon (M-series) via PyTorch MPS. All training and evaluation runs fit within a \$50 compute budget; in practice the project consumed zero paid compute (all on local hardware over ~ 24 hours of background time).

4 Single-SAE Results (Baseline)

4.1 Reconstruction and sparsity

On 50,000 held-out activations from the training-document set: variance explained 98.43%, mean $L_0 = 63.9 \pm 1.1$ (target $k = 64$, exact), reconstruction MSE summed over d_{model} is 25.74. Dead features (features that never activate): $855/16{,}384 = 5.22\%$.

4.2 CE-delta evaluation

Following [6], we evaluate the SAE by substituting its reconstruction into the language model's forward pass at the trained layer, and measuring the increase in downstream cross-entropy on held-out tokens. For 100,000 held-out tokens:

$$\text{CE}_{\text{clean}} = 3.107 \text{ nats/token} \tag{4}$$
$$\text{CE}_{\text{SAE-patched}} = 3.383 \text{ nats/token} \tag{5}$$
$$\text{CE}_{\text{zero-ablate (mean)}} = 9.122 \text{ nats/token} \tag{6}$$

The SAE-patched run substitutes $\hat{h} = \text{SAE}(h)$ for h at layer 6; the zero-ablation run substitutes the data mean (b_{dec}). The loss-recovered metric is $1 - \Delta\text{CE}/(\text{CE}_{\text{ablate}} - \text{CE}_{\text{clean}}) = 95.4\%$, in the strong end of the published small-model band [1, 6].

4.3 Auto-interpretation catalog

We extract top-10 max-activating snippets per feature on the 1M- token training stream, then submit the top 26 features (ranked by a mixture of peak and mean-of-top-K activation) to Kimi-K2 (`moonshot-v1-32k`) for auto-interpretation. Each receives a one-sentence concept label and a classification of MONOSEMANTIC, POLYSEMANTIC, or UNCLEAR. Total API cost: \$0.02.

Of the 26:

- MONOSEMANTIC: 23 (88.5%).
- POLYSEMANTIC: 3 (11.5%).
- UNCLEAR: 0.

Representative monosemantic features (full table in Appendix B): *Decimal points in numerical contexts* (f1989), *Exponentiation notation in mathematical contexts* (f5196), *Tokens related to file paths and directory structures* (f5747), *Newline tokens inside code blocks* (f7448), *BibTeX/LaTeX citation references* (f12117), *Logical operators and negation symbols* (f12697).

Apparent feature splitting. Approximately 11 of the 23 monosemantic features received label variants of "newline tokens": *newline tokens in various contexts*, *newline tokens inside XML/HTML/CSS blocks*, *newline tokens inside code blocks*, and so on. The SAE has clearly split a single semantic category into many features. The labeling LLM identifies each as monosemantic but cannot give distinct labels because the splits are subtle. This is direct evidence for feature splitting at $N = 16{,}384$ on this model.

5 Cross-Validation Results (the New Material)

5.1 Cross-width sweep

Three SAEs trained at $N \in \{4{,}096, 16{,}384, 65{,}536\}$ with all other hyperparameters held fixed. Results in Table 1.

Table 1: Cross-width Pareto. Wider SAEs do not buy meaningful reconstruction beyond $N \approx 4\text{k}$ on this data scale, and dead-feature counts explode at $N = 65{,}536$.

Width N	MSE	Var. explained	Dead features	Alive features
4,096	33.11	0.9782	1 (0.02%)	4,095
16,384	25.74	0.9843	855 (5.22%)	15,529
65,536	23.68	0.9853	41,698 (63.63%)	23,838

Two observations. **The Pareto curve in width is nearly flat above** $N = 4{,}096$: going 4k $\to$ 64k ($16\times$ more capacity) buys 0.71 percentage points of variance explained. **Dead features explode at** $N = 65{,}536$: 63.6% of features never fire on any input.

Feature merging at small width. We find one 4k feature (f2175) is the best cosine-similarity match for five different 16k newline-variant features (cos 0.34 to 0.39). Smaller widths collapse multiple features that nominally differ at larger widths.

Feature splitting at large width does not survive seed change. Anticipating §5.2: most of the apparent 16k features that look like "finer-grained splits" of 4k features do not have stable counterparts in a second random-seed SAE.

5.2 Cross-seed replication

We train a second SAE at identical hyperparameters with `torch.manual_seed(42)` (the reference SAE used the default implicit seed). The two SAEs have statistically indistinguishable summary metrics: variance explained 98.43% vs. 98.44%, dead features 5.22% vs. 5.47%, $L_0 = 63.9$ for both.

For each feature in the reference, we find its best-match decoder column in the second SAE under cosine similarity (with decoder columns unit-norm across both). Distribution of best-match cosine similarities:

Threshold	Count above	Fraction
cos > 0.50	12,892	0.787
cos > 0.70	9,783	0.597
cos > 0.90	3,254	**0.199**
cos > 0.95	952	0.058
cos > 0.99	14	0.001

Only 19.9% **of features have a cross-seed match above cos** $= 0.9$**.** Same model, same data, same hyperparameters, same architecture: only 1 in 5 features replicates.

All 11 **"newline-variant" features in the reference SAE collapse to a single feature in the seed-**42 **SAE.** Specifically, the best-match feature in the seed-42 SAE for each of the labeled newline features is the same feature (f12024) with cosine similarities of 0.34–0.46. This directly shows that the apparent "feature splitting" observed at $N = 16{,}384$ is substantially a training-noise artifact rather than the SAE discovering finer-grained concepts.

5.3 Cross-architecture: Matryoshka vs. flat TopK

We train a Matryoshka SAE [7] on the same 1M activations with $k_{\text{levels}} = \{16, 64, 256\}$ and the same $N = 16{,}384$. Per-level variance explained: $\{0.9747, 0.9830, 0.9889\}$. The nested-subset invariant (features at k_i are a strict subset of features at k_{i+1}) holds for 100% of evaluated tokens by construction.

Compared to the flat TopK reference at $k = 64$: the Matryoshka at the $k = 64$ level has slightly worse reconstruction (0.983 vs. 0.984 variance explained) but dramatically fewer dead features (1.64% vs. 5.65%). The multi-level training appears to keep more capacity alive.

Decoder column cosine similarity across architectures:

Threshold	Count above	Fraction
cos > 0.50	12,781	0.780
cos > 0.70	9,517	0.581
cos > 0.90	1,745	**0.106**
cos > 0.95	144	0.009
cos > 0.99	4	0.0002

Cross-architecture stability is even worse than cross-seed: 10.6% of features at cos > 0.9, vs. 19.9% cross-seed within the same architecture. Matryoshka and flat TopK find substantively different feature bases on the same model and the same data.

5.4 Joint stability across all conditions

For each of the 16,384 features in the reference SAE, we count the number of cross-validation conditions in which it has a best-match cosine > 0.9. The conditions are: 4k-width SAE, 64k-width SAE, seed-42 SAE, Matryoshka SAE. Maximum possible score: 4.

Table 2: Joint stability across all four cross-validation conditions. Only 2.14% of features survive every comparison at cos > 0.9.

Stability score (cos > 0.9)	# features	Fraction
0/4	11,974	73.08%
1/4	1,799	10.98%
2/4	1,272	7.76%
3/4	988	6.03%
4/4	**351**	**2.14%**

The 351 fully stable features are the strongest "real feature" candidates in this setup — their decoder direction replicates regardless of width, seed, or architectural objective.

Of the 26 auto-labeled monosemantic features, only 3 are fully stable:

- f12117 — BibTeX/LaTeX citation references (cos ≥ 0.997 across every comparison)
- f5747 — File paths and directory structures (cos ≥ 0.997 across every comparison)
- f12697 — Logical operators and negation symbols (cos ≥ 0.977 across every comparison)

The remaining 23 labeled features have stability score $\leq 1/4$: they are clearly identified by auto-interp but do not survive any non-trivial replication test. We take this as direct evidence that **auto-interpretation overstates feature reliability by a factor of approximately** $8\times$ on this setup (88.5% monosemantic vs. 11.5% fully stable).

5.5 Multi-metric scorecard

The six axes evaluated on the reference SAE:

Axis	Value	Source
Reconstruction (var. explained)	**0.984**	basic eval
Predictive preservation (CE-delta loss recovered)	**0.954**	§4.2
Capacity utilisation (1− dead fraction)	**0.948**	basic eval
Auto-interp monosemanticity rate	0.885	§4.3
Cross-seed stability (cos > 0.9)	**0.199**	§5.2
Cross-architecture stability (cos > 0.9)	**0.106**	§5.3

The range is 0.106 to 0.984, a ~ 88 percentage-point gap. The six axes do not agree. Reporting only the strongest axis (variance explained) overstates the SAE's quality by nearly an order of magnitude relative to the weakest axis (cross-architecture stability).

6 Discussion

6.1 The standard suite is necessary but not sufficient

Reconstruction, sparsity, and monosemanticity-rate are useful but incomplete. They measure how well the SAE can encode and decode a given model's activations and how interpretable the recovered features look. They do not measure whether the features are a property of the model or an artifact of the SAE training run. Our cross-validation results indicate that, on this setup, the standard suite overstates feature reliability by $\sim 8\times$.

The implication is not that SAEs are useless — the 3 fully stable features we recover are clean, meaningful, and would survive any reasonable interpretability test. The implication is that without replication-style metrics, an SAE quality report cannot be cited as evidence about the underlying model.

6.2 What is the field's current practice?

Most published SAE work reports reconstruction-style metrics on a single trained SAE. A growing minority reports CE-delta substitution [6]. Cross-seed replication is occasionally acknowledged as a limitation [2, 4] but rarely measured quantitatively at the feature level. Cross-architecture replication (different SAE objectives on the same model) is, to our knowledge, not standard practice.

We argue that cross-seed replication is the cheapest improvement available — one extra training run, identical hyperparameters — and should be reported alongside reconstruction in any SAE quality claim.

6.3 The shortlist of stable features as a basis for downstream work

The 351 fully stable features (2.14% of the reference SAE) form a small, well-validated candidate set for downstream interpretability work — circuit discovery, causal interventions, agent-behavior auditing. Building circuits on the full SAE risks finding patterns that are themselves training-noise artifacts.

6.4 Tools we release

1. `11_sae_scorecard.py` — given a trained SAE checkpoint and the standard set of evaluation JSON outputs, prints the six-axis quality profile (§5.5).

2. `12_feature_navigator.py` — given a reference SAE and any set of comparison SAEs, computes per-feature stability scores and emits both a JSON for programmatic use and a Markdown shortlist of fully stable features. Supports `--query <feature_id>` for single-feature lookup.

7 Limitations

- **Single layer, single model.** We evaluate one residual stream at one layer of one 160M-parameter model. The numerical claims of 2.14% joint stability or $8\times$ overstatement should not be read as universal. The methodological claim — that reconstruction-style and replication-style metrics can disagree by an order of magnitude — is more transportable.

- **Single random seed pair.** Cross-seed stability is measured across exactly two SAEs (reference and seed-42). Stability across more seeds may yield different (likely lower) numbers.

- **No causal validation.** We do not perform feature ablations to test that fully stable features are also causally load-bearing for downstream behavior. Cross-validation stability is a cheap proxy, not a substitute for causal validation [9, 10].
- **Auto-interpretation is partial.** The Kimi-K2 labeling collapses fine-grained splits we know exist (e.g., 11 newline-variant features all labeled "newline tokens in various contexts"). A stronger labeling model would likely raise the monosemanticity rate further without changing the cross-validation stability, which would widen the reconstruction-vs.-replication gap.
- **Computational match.** A linear-encoder TopK SAE is computationally matched to a single MLP layer; this constrains the class of features it can find. Stronger SAE variants (multi-layer encoders, attention-based encoders) could fit features the model itself cannot use, exacerbating the cross-validation gap.
- **1M tokens is small.** Modern frontier-scale SAE work uses billions of activations. Some quantitative findings (e.g., the $N \approx 4$k effective capacity, the 63% dead-feature rate at $N =$ 65,536) may shift at larger training scales.

8 Conclusion

We evaluated a TopK SAE on Pythia-160M layer 6 along six independent quality axes including three cross-validation tests. The standard reconstruction-style metrics (variance explained, CE-delta loss recovered) and the standard interpretability metric (auto-interp monosemanticity rate) all score the SAE highly — between 0.88 and 0.98. The replication-style metrics (cross-seed, cross-architecture, cross-width consistency) score the same SAE between 0.10 and 0.20. The gap is large enough that the standard suite cannot be the only quality claim attached to an SAE: it overstates feature reliability by roughly 8× on this setup.

We release two tools — a multi-metric scorecard and a per-feature stability navigator — that surface this gap on any SAE. We expect cross-seed replication to become standard in SAE reports; the marginal cost is one extra training run, and the marginal information is substantial.

Code and Artifacts

https://github.com/OE-GOD/sae-pythia-160m

All checkpoints regenerable via the scripts in `code/`; raw analysis outputs in `data/`; the trained 16k reference SAE, 4k and 64k width SAEs, seed-42 SAE, and Matryoshka SAE are described and reproducible via the README.

Acknowledgments

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A The L1 Shrinkage Failure Mode We Started With

Before switching to TopK, we trained a vanilla L_1 SAE on the same setup and swept $\lambda \in \{0.005, 0.02, 0.1\}$ at $N = 16{,}384$, 1500 training steps each. Despite a 20× increase in sparsity-penalty strength, the average L_0 remained between 1,048 and 1,333 active features per token — well above the target band of 30-100.

This is the documented "L_1 shrinkage" failure mode [4]: minimising $\lambda \sum_i |f_i|$ is indifferent between "one feature firing at strength S" and "S features firing at strength 1". Switching to TopK fixes this by hard construction; $L_0 = k$ becomes exact.

Table 3: L_1 SAE sparsity sweep.

λ	Final MSE	Var. explained	L_0
0.005	2.80	0.998	1333
0.02	5.20	0.997	1302
0.1	14.72	0.991	1048

B Selected Auto-Labeled Features

The full 26-feature catalog is in `data/feature_catalog.json`. Representative monosemantic features:

ID	Auto-interpretation
f1989	Decimal points in numerical contexts
f5196	Exponentiation notation in mathematical contexts
f5747	Tokens related to file paths and directory structures
f7448	Newline tokens inside code blocks
f10045	Subword BPE continuations of multi-piece words
f12117	BibTeX/LaTeX citation references
f12697	Logical operators and negation symbols

Fully stable across cross-validation (cos > 0.9 in all 4 comparisons): f12117, f5747, f12697.